

Test

1. Java is a programming language used to build an applications.

Example: mobile applications, Desktop applications ,etc.

Java is called as platform independence because it can run a code in any software by the use of JVM
It follows WORA means write once and run anywhere.

2. JDK is called java development kit

It helps to build a complete java applications

JDK as 2 parts called

- Tools
- JRE(Java runtime environment)

Tools as javac which helps to compile the code and convert it into bytecode (.Class) which is called intermediate code.

JRE is called java run time environment which helps to run the java code

JRE as 2 parts called

- Library
- JVM

Lib is the folder which as some predefined files in it.

JVM it is called java virtual machine which converts the byte code into machine code in the binary form and then helps the code to run in secure way

3. JVM

JVM is called java virtual machine which helps to convert byte code into machine code and helps to run the code.

JVM is platform dependent because each operating system as its own processors, memory and many other things so JVM must need to integrate for every system for better work.

JVM helps to keep the code secure.

4. WORA

WORA is all about write once and run anywhere here it is called platform independence because the code which is written once or the application using java which is done once it can be run in any operating system without any problem it is very secure.

5. Bytecode

In java .class is called bytecode below flow will show the process

MyPro.java -> javac -> Bytecode -> output

Here the file name is called MyPro which is saved in .java format then the javac tool will help to compile the code it convert .java to .class which is called byte code .

Byte code is not machine code and not a human code it is an intermediate code and it is platform independent when the bytecode is generated it is converted into machine code with the help of JVM and JVM will run the program .

6. Datatype

Datatype specifies the type or a range the assigned value can hold.

Datatypes are of 2 types:

- Primitive datatype
- Non-primitive datatype

Primitive datatypes:

Primitive datatypes are the types which as the predefined contents which was declared by java

There are 8 primitive datatypes:

Int
Float
Byte
Short
Char
Boolean
Long
Double

Non-primitive datatypes:

Non-primitive datatypes are datatypes which are created by programmers.

There are infinite non-primitive datatypes.

Some of them are: String, array, object etc.,

7. Default values

Default values are the types which are accessed within the same package or in the sub-class in sub-class there may be different packages.

```
class AllTypes{
static void values(){
String name="john";
int num=4;
float leng=3.5f;
double width=2.345;
short numbr=2;
long total=23456L;
boolean isCorrect=true;
System.out.println("Name:"+name);
System.out.println("num:"+num);
```

```
System.out.println("Length:"+leng);
System.out.println("width:"+width);
System.out.println("number:"+numbr);
System.out.println("total:"+total);
System.out.println("is correct:"+isCorrect);
```

```
}
}
```

```
class AllTypeRunner{
public static void main(String[] args){
AllTypes.values();
}
}
```

OUTPUT:

C:\Learning java\newone>java AllTypeRunner

Name:john

num:4

Length:3.5

width:2.345

number:2

total:23456

is correct:true

8. Variables

Variables are the containers to store the values.

Static variables are the variables which belongs to class.

Example:

```
1.class Bus{
    static color="red";
static void details(){
System.out.println("it is a bus");
System.out.println(color);
}
```

Local variables are the type of variables which are declared within the methods or a constructors.

Example:

```
class Bag{
    static void details(){
        string color="red";
        int weels=4;
    }
```

9. Static method

Static methods are the non-accesses modifiers which can have method and a variables and the static methods are mainly identified as it belongs to class not for an object.

Syntax to call static method

```
className.methodName();
```

In order to invoke the static method we must need to right the name of the class and the name of the methos and the parameters are optional within the paranthesis and ends with semicolon.

The names of the class and methods are case sensitive.

10.

```
class Shopekeeper{
static String[] details(){
    int count=0;
    int[] sales={10,20,30,40,50};
    for(int start=0;start<sales.length;start++){
        count+=sales[start];
    }
    System.out.println("the total count:"+count);
    System.out.println("The higest value is:"+sales[4]);
    return null;
}
}
class shop{
public static void main(String[] args){
    Shopekeeper.details();
}
}
```

OUTPUT:

the total count:150

The higest value is:50